

A Layered Model for Generative AI–Mediated Early Requirements Engineering in Digital Government

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Abstract. Early Requirements Engineering (Early RE) faces challenges related to ambiguity, informal stakeholder communication, and reliance on tacit knowledge—issues that are exacerbated in digital government contexts. While Large Language Models (LLMs) have shown promise in supporting isolated RE tasks, there is still no integrated model that defines how Generative Artificial Intelligence (GenAI) should participate across the entire Early RE lifecycle. Existing approaches tend to apply GenAI uniformly as a text generator, overlooking the distinct communication dynamics of each phase.

This paper proposes a layered model for GenAI-mediated Early RE grounded in three design principles: phase-specific role differentiation, a gradient of human control, and risk awareness as a cross-cutting concern. The model comprises four functional layers—*Mediator* during elicitation, *Co-creator* during analysis, *Facilitator* during negotiation, and *Assistant* during early validation—alongside a transversal risk management layer addressing hallucinations, biases, and context loss.

We define the behavior of each layer, the flow of artifacts, and the interactions between actors, and illustrate the model through a detailed scenario in municipal digital government. Finally, we discuss implementation considerations for developing phase-aware GenAI-mediated RE tools.

Keywords: Requirements Engineering · Generative AI · Layered Model · LLM · Digital Government

1 Introduction

The early phases of Requirements Engineering (RE)—elicitation, analysis, negotiation, and validation—are critical for determining system scope [1,2]. They are challenging as stakeholders express needs informally, requiring translation into structured artifacts and reconciliation of priorities [3]. These challenges peak

in digital government projects, where officials use bureaucratic terms, citizens use colloquial language, and requirements must satisfy complex regulatory constraints [4].

While Large Language Models (LLMs) show promise for these challenges via natural language understanding and reasoning [5], their application in RE has been predominantly task-oriented (e.g., generating stories, detecting ambiguities) without considering how Artificial Intelligence (AI) participation should vary across phases [6]. This uniform application fails to exploit LLMs’ full potential for mediation, conflict analysis, and verification.

This article proposes a layered model for AI-mediated Early RE, introducing *role differentiation by phase*. It defines a specific role for each phase calibrated to its communication dynamics, formality, and artifacts. A cross-cutting risk management layer addresses threats introduced by Generative AI (GenAI).

The article is organized as follows. Section 2 provides background. Section 3 presents the design principles. Section 4 describes the model architecture. Section 5 illustrates a scenario. Section 6 discusses implementation. Section 7 presents related work. Section 8 discusses strengths and limitations. Section 9 concludes.

2 Background

2.1 Early Requirements Engineering

Following standard frameworks [1,2], Early RE encompasses four iterative phases:

Elicitation captures needs from stakeholders. The main challenge is the communication gap between stakeholders and engineers, which amplifies in domains like digital government [4].

Analysis structures and formalizes elicited information into RE artifacts (e.g., user stories, models), managing ambiguity and inconsistencies.

Negotiation resolves conflicts and prioritizes demands to reach consensus, a complex socio-political process [3].

Early Validation verifies that artifacts reflect stakeholder intentions, are consistent, and traceable.

These phases interact iteratively. Findings in validation may trigger new elicitation or re-analysis.

2.2 GenAI/LLMs in Requirements Engineering

Recent reviews [5,6] document LLMs applied to RE tasks like generating user stories [7], classification, and ambiguity detection. These applications share a pattern: AI acts as a task executor producing text from a prompt. What is absent is a systematic consideration of how AI’s role should vary throughout the RE process. Current approaches treat AI as a uniform resource, ignoring the different dynamics of each phase.

3 Design Principles

The model is grounded in three design principles derived from analyzing Early RE dynamics and current LLM application characteristics.

3.1 P1: Role Differentiation by Phase

Each Early RE phase involves different actors, communication patterns, and artifact types. Elicitation is mainly about understanding (bridging the stakeholder-engineer gap); analysis is about construction (crafting formal artifacts from informal inputs); negotiation is about resolution (finding compromises among conflicting interests); validation is about verification (systematically checking produced artifacts).

A model assigning the same AI role to all phases—typically “generator”—misses these distinctions. AI should behave differently depending on the phase: more proactive in bridging communication during elicitation, more collaborative in co-constructing artifacts during analysis, more neutral in presenting compromises during negotiation, and more constrained in executing systematic checks during validation.

3.2 P2: Human Control Gradient

The degree of human oversight should increase as the process advances toward commitment. In elicitation, where AI bridges communication, some autonomy is acceptable—AI can rephrase, summarize, and suggest without explicit approval for every micro-action. In validation, where AI verifies artifacts that will drive downstream design decisions, it should operate strictly under the analyst’s direction, executing only requested checks.

This gradient reflects both practical necessity (elicitation involves too many micro-interactions for per-action approval) and risk management (validation outputs have more direct downstream impact than elicitation drafts).

3.3 P3: Risk Awareness as a Cross-Cutting Concern

Generative AI introduces specific risks into the RE context that are distinct from general hallucination or bias concerns:

- **Hallucinated requirements:** AI might generate plausible-sounding requirements that no stakeholder actually expressed, or falsely attribute needs to specific stakeholders.
- **Domain bias:** AI might reproduce patterns from its training data that do not reflect the local context (e.g., assuming standard workflows that do not apply to a specific municipal government’s processes).
- **Context loss:** In long elicitation or analysis sessions, AI may lose coherence, generating artifacts inconsistent with earlier parts of the same session.

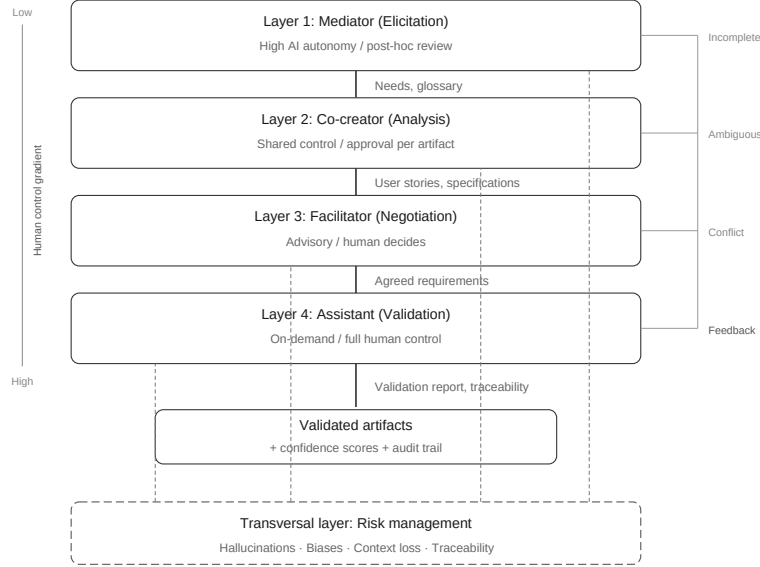


Fig. 1. Layered model for AI-mediated Early RE. The four functional layers are organized sequentially with artifact flow between them. The human control gradient increases from top (high AI autonomy) to bottom (full human control). The cross-cutting risk management layer operates across all functional layers. Dotted arrows represent feedback loops.

- **Traceability fabrication:** AI might generate traceability links that seem valid but do not correspond to actual relationships between requirements and their sources.

These risks are not specific to any single phase; they can manifest in any layer. Therefore, risk management must be a cross-cutting concern, not a per-phase ad hoc mechanism.

4 The Layered Model

The model defines four functional layers—one per Early RE phase—plus a cross-cutting risk management layer. Each layer specifies the AI’s role, its key behaviors, the artifacts it consumes and produces, and its position on the human control gradient. Figure 1 presents the overall architecture of the model.

4.1 Layer 1: Mediator (Elicitation)

Role: Communication bridge between non-technical stakeholders and developers. **Behaviors:** Translate informal language to semi-structured candidates, ask

clarifying questions, build a domain glossary, detect implicit needs, and summarize sessions. **Artifacts (In → Out):** Transcripts/docs → Raw candidates, glossary, clarification log. **Control:** *High AI autonomy / post-hoc review*. AI drafts autonomously; analyst reviews after generation.

4.2 Layer 2: Co-creator (Analysis)

Role: Active participant in constructing formal RE artifacts alongside the analyst. **Behaviors:** Generate user stories and acceptance criteria, detect ambiguities/incompleteness, suggest missing requirements (gap analysis), and propose domain models. **Artifacts (In → Out):** Raw candidates/glossary → User stories, acceptance criteria, ambiguity flags, gap report. **Control:** *Shared control / per-artifact approval*. AI proposes; analyst decides on each generated artifact.

4.3 Layer 3: Facilitator (Negotiation)

Role: Objective support for negotiation without making decisions. **Behaviors:** Identify conflicts, quantify trade-offs, present neutral resolution options, suggest compromise strategies, and document decisions. **Artifacts (In → Out):** Approved artifacts/priorities → Conflict matrix, trade-off analysis, compromise proposals, agreed requirements. **Control:** *Advisory / human decides*. AI presents options; humans make all decisions.

4.4 Layer 4: Assistant (Early Validation)

Role: Executes systematic verification under explicit analyst direction. **Behaviors:** Verify consistency, check completeness against templates, generate traceability links, flag residual ambiguities, and produce validation reports. **Artifacts (In → Out):** Agreed requirements/accumulated artifacts → Validation report, traceability matrix, issue list. **Control:** *On-demand / full human control*. AI executes only requested checks.

4.5 Cross-Cutting Layer: Risk Management

Role: Continuous monitoring and mitigation of AI-specific risks across layers. **Behaviors:** Attach confidence scores to AI-generated artifacts, flag potential hallucinations (untraceable claims), monitor context coherence, verify traceability integrity, and maintain an audit trail. **Scope:** Operates across all layers, attaching metadata to each artifact. **Outputs:** Risk alerts, confidence scores, coherence warnings, audit trail.

4.6 Inter-Layer Interactions and Artifact Flow

The layers form a pipeline with explicit artifact flow:

1. The *Mediator* (Layer 1) produces raw requirements and a glossary that feed the *Co-creator* (Layer 2).
2. The *Co-creator* produces structured artifacts (user stories, acceptance criteria) that may reveal conflicts, triggering the *Facilitator* (Layer 3).
3. The *Facilitator* produces agreed requirements that the *Assistant* (Layer 4) validates.
4. Validation findings may trigger *feedback loops*: incomplete requirements return to elicitation (Layer 1), ambiguous artifacts return to analysis (Layer 2), unresolved conflicts return to negotiation (Layer 3).

The cross-cutting Risk Management layer operates continuously, enriching each artifact produced by any layer with confidence scores and risk flags. Subsequent layers can use this metadata to treat low-confidence artifacts with appropriate caution.

5 Illustrative Scenario

We illustrate the model’s operation through a scenario based on a municipal digital government context: the development of a citizen complaint management system for the General Directorate of Modernization of a municipal government in Argentina.

5.1 Elicitation (Mediator)

A municipal official from the Citizen Service area is interviewed. She describes the current manual process:

“When someone comes to complain about a pothole or a broken streetlight, we fill out a paper form. Then we take the form to Public Works. Sometimes weeks pass before anything happens, and the citizen calls back asking what’s going on, and we have no way to tell them.”

The Mediator layer processes this input and produces:

- **REQ-01:** The system shall allow citizens to register infrastructure complaints.
- **REQ-02:** The system shall notify the Public Works department when a complaint is registered.
- **REQ-03:** The system shall track the resolution status of complaints.
- **REQ-04:** The system shall allow citizens to check the status of their complaints.
- **Glossary entry:** “Complaint” = report initiated by a citizen about a municipal infrastructure issue requiring intervention.
- **Clarifying question:** “Should citizens be able to attach photographs of the issue?”

The Risk Management layer flags REQ-04 as a potential inference: the official mentioned that citizens “call back” to ask for status, but did not explicitly request a self-service query feature. Confidence score: 0.7.

5.2 Analysis (Co-creator)

The Co-creator layer takes the raw requirements and generates structured artifacts:

User Story: *“As a citizen, I want to submit a complaint with a description and an optional photo, so that the municipality can identify and address the issue.”*

Acceptance Criteria:

- The complaint form must include: location, category (pothole, lighting, drainage, other), free-text description, and optional photo upload.
- Upon submission, the citizen receives a confirmation with a tracking number.
- The Public Works department receives a notification within 5 minutes of submission.

Ambiguity flag: REQ-02 specifies “notify” but does not define the notification channel (email, dashboard alert, SMS, push notification).

Gap detection: No requirement addresses complaint prioritization, response time commitments, or escalation procedures.

5.3 Negotiation (Facilitator)

The Public Works department requests that complaints be automatically prioritized by severity (safety-critical first). The Citizen Service area insists on chronological processing (first-come, first-served) to ensure perceived fairness.

The Facilitator layer identifies this as a **priority conflict** and presents the analysis:

- **Option A (Severity-based):** Reduces average resolution time for critical issues (estimated 40%) but may delay low-severity complaints.
- **Option B (Chronological):** Perceived as fair by citizens but may leave safety-critical issues unaddressed for longer.
- **Option C (Hybrid):** Two-tier system—safety-critical complaints are prioritized regardless of submission order; all others follow chronological order.

The AI does not recommend an option; the stakeholders choose Option C after discussion.

5.4 Early Validation (Assistant)

The analyst requests a completeness and consistency check. The Assistant identifies:

- REQ-03 (track resolution status) lacks a corresponding user story.
- Acceptance criteria for photo uploads do not specify maximum file size or accepted formats.
- The notification mechanism for REQ-02 remains undefined after negotiation.
- Non-functional requirements (performance, availability, accessibility) have not been specified.

The traceability matrix links each requirement to its source (line number of the interview transcript) and its derived artifacts (user stories, acceptance criteria).

6 Implementation Considerations

The model is designed to be LLM-agnostic: it specifies *what* the AI should do in each phase without prescribing *which* model to use. However, implementation requires several architectural decisions:

Per-layer prompt engineering. Each layer requires a distinct system prompt defining the AI’s role, behavioral constraints, and output format. The Mediator prompt emphasizes empathetic listening and vocabulary bridging; the Co-creator prompt emphasizes structured output and gap detection; the Facilitator prompt emphasizes neutrality and trade-off analysis; the Assistant prompt emphasizes systematic verification.

Context management. LLMs have finite context windows. Long elicitation sessions or large requirement sets may exceed this window, requiring strategies like summarization, chunking, or Retrieval-Augmented Generation (RAG) to maintain coherence throughout the session.

Confidence scoring. The Risk Management layer requires a mechanism to assign confidence scores to AI-generated artifacts. This can be implemented through calibrated uncertainty estimation, source attribution tracking (which parts of the output are directly grounded in stakeholder input vs. inferred by the AI), or a combination of both.

Artifact persistence and traceability. All artifacts must be stored with their provenance metadata: which layer generated them, from what inputs, with what confidence score, and whether they have been human-approved. This enables the audit trail required by the Risk Management layer.

7 Related Work

Several lines of research inform and contrast with our proposal.

LLMs in RE. Arora et al. [7] demonstrated that GPT-4 can generate user stories from natural language descriptions with reasonable quality, but their work addresses only the elicitation-to-analysis transition without a broader process model. Ronanki et al. [8] explored ChatGPT for requirements tasks in an exploratory study, covering elicitation and analysis but without role differentiation or risk management.

AI Risks in Software Engineering (SE). Zhang et al. [9] and Ji et al. [10] provide comprehensive taxonomies of hallucinations that inform our cross-cutting layer. However, neither work addresses the specific manifestation of hallucination in RE, where a fabricated requirement has different consequences than a fabricated code snippet or factual claim.

RE process models. Classic RE process models [1,2] define the phases we adopt but predate the AI era. Our model extends these frameworks by defining AI participation within each phase, rather than replacing the human-centric process.

Prompt engineering for SE. White et al. [11] proposed a catalog of prompt patterns for software engineering. Their patterns (e.g., persona pattern,

template pattern) can be used to implement the per-layer prompts in our model, but do not address the question of how prompts should vary across RE phases.

Our model is, to our knowledge, the first to propose role differentiation per RE phase with an explicit human control gradient and a cross-cutting risk management mechanism.

8 Discussion

8.1 Strengths

The model offers three main strengths. First, role differentiation provides a grounded basis for designing AI-mediated RE tools that behave differently depending on the phase, exploiting LLM capabilities beyond simple generation. Second, the human control gradient explicitly addresses concerns about AI autonomy in critical engineering activities, ensuring AI influence is most constrained where decisions have the highest downstream impact. Third, the cross-cutting risk layer treats AI reliability as an architectural concern, enabling systematic rather than ad hoc risk management.

8.2 Limitations

The model has several limitations that must be acknowledged. The boundaries between Early RE phases are not always sharp in practice; agile and iterative processes can blur the distinction between elicitation and analysis, or between negotiation and validation. Feedback loops partially address this, but empirical validation is needed to assess how the model handles fluid process boundaries.

The model does not address late-phase RE activities (requirements management, evolution, change management), which involve different dynamics. Extending the model to cover the full RE lifecycle is a potential direction for future work.

The effectiveness of each AI role depends on the capabilities of the underlying LLM, which vary across models and evolve rapidly. The model’s LLM-agnostic design partially mitigates this, but performance differences between models may require layer-specific model selection.

8.3 Target Validation Domain

The model targets empirical validation in the General Directorate of Modernization of the Municipal Government of San Fernando del Valle de Catamarca, Argentina. This digital government context is particularly suitable because it combines non-technical stakeholders, regulatory constraints, multi-area coordination requirements, and active software development projects for the digitalization of municipal services.

9 Conclusion

This article proposed a layered model for AI-mediated Early Requirements Engineering that assigns differentiated roles to AI across the four Early RE phases: mediator in elicitation, co-creator in analysis, facilitator in negotiation, and assistant in early validation. A cross-cutting risk management layer addresses hallucinations, biases, context loss, and traceability fabrication as cross-cutting concerns.

The model is grounded in three design principles—role differentiation, human control gradient, and risk awareness—and provides a conceptual foundation for building AI-mediated RE tools with phase-appropriate behavior. An illustrative scenario in municipal digital government demonstrated the model’s applicability and the distinct contributions of each layer.

Ongoing work focuses on developing a functional prototype that implements the model using LLM Application Programming Interfaces (APIs) with phase-specific prompting strategies, and its empirical validation in real digital government projects.

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